Data Analyst Jobs in India

Exploratory Data Analysis of live job postings

Data source: Adzuna Jobs API (India) | Tools: Python, pandas, numpy, matplotlib, seaborn

Report date: 13 October 2026 | Postings collected: 10 October 2026

161	**57**	**7.5 LPA**	**26%**
data analyst jobs analysed	jobs with a usable salary	median listed salary	of jobs ask for SQL

1. Executive summary

This project studies the **Data Analyst** job market in India using live postings from the Adzuna API, so that a student can see what employers actually ask for. After collecting 2,675 IT job postings and 50 dedicated data analyst postings, and keeping only titles that contain "data analyst", the final dataset has **161 jobs**.

Key findings

- **Location:** Bangalore has 53 of the 126 jobs that name a city (42%). Pune is second (13) and Noida third (9). Delhi NCR (Noida + Gurugram + Delhi) adds up to 19.
- **Companies:** hiring is very spread out: 137 companies for 159 jobs. The top company has only 4 jobs. Some are recruitment platforms.
- **Experience:** where stated (35 jobs), the median minimum is **2 years**. 18 jobs ask for 0-2 years, 14 for 3-5 years and only 3 for 6 or more.
- **Salary:** median **7.5 LPA**; the middle half of jobs pay 6 to 11 LPA (mean 9.7, highest 32.4). Bangalore's median (9.5) looks higher than other cities (7.25), but the groups are small.
- **Tools:** SQL is asked most (26%), then Python (16%), Power BI and Excel (about 11% each) and Tableau (7.5%). SQL + Python is the strongest pair.
- **The work itself:** insights/trends (32%), reporting (29%) and dashboards (15%) appear far more than machine learning (3%).
- **Other:** only 4% mention remote or hybrid work; 59% of postings are from 2026 and 33% are from the last 30 days.

> **For a beginner (this sample only):** start with SQL, add Python (pandas) and one BI tool (Power BI or Tableau), and practise building dashboards and explaining insights.

2. Aim and questions

The goal is to understand the data analyst job market in India from live postings. Five questions guide the analysis:

- **Where** are data analyst jobs, and **which companies** are hiring?
- How much **experience** do employers ask for?
- How much do these jobs **pay**, and what changes the salary (experience, city, tools)?
- Which **tools** are asked for most, and which tools come together?
- Are there **remote** jobs, and how fresh are the postings?

3. Data collection

A Python script (`fetch_jobs.py`) calls the Adzuna India API with 18 search terms (for example "data analyst", "software developer", "data engineer", "business intelligence analyst"), category "it-jobs", 50 results per page and up to 3 pages per term, with a 1-second pause between calls. Duplicate job ids are skipped and the result is saved as `data/it_jobs_india.csv`. API credentials are read from a `.env` file and kept out of version control through `.gitignore`.

File	Rows	Content
it_jobs_india.csv	2,675	All IT jobs from 18 searches
data_analyst_jobs.csv	50	Dedicated data analyst search (different column names, renamed to match)
data_analyst_jobs_clean.csv	161	Final cleaned dataset with 49 columns, including engineered features

4. Method

Selection and cleaning

- Joined both files, removed repeated job ids, and kept titles containing "data analyst" (161 jobs). 65 BI-analyst and 9 business-analyst jobs were left out on purpose.
- Removed postings with the same title, company, location and description.
- Parsed dates; created year and days-old columns.
- Kept only the city from the location ("Bengaluru" mapped to Bangalore, "Gurgaon" to Gurugram); "India" alone became "Not specified".
- Salary is yearly in rupees. A salary was kept only if both minimum and maximum were at least Rs 1,00,000, which removes wrong values (0, 6) and monthly figures. It was converted to LPA (lakhs per annum) using the midpoint.

Feature engineering

- **Seniority** from title words (senior, lead, junior, fresher ...).
- **Minimum experience** extracted from phrases such as "3+ years" or "2-5 years" (values above 15 discarded), then grouped into bands.
- **Remote/hybrid** flag from the words remote, work from home, WFH or hybrid.
- **16 tool flags** (SQL, Excel, Power BI, Tableau, Python, R, Looker, BigQuery, Snowflake, Azure, AWS, GCP, Spark, ETL, SAS, SPSS) and **9 concept flags** (dashboards, reporting, statistics, machine learning ...), found with text patterns.

5. Findings

5.1 Where are the jobs?

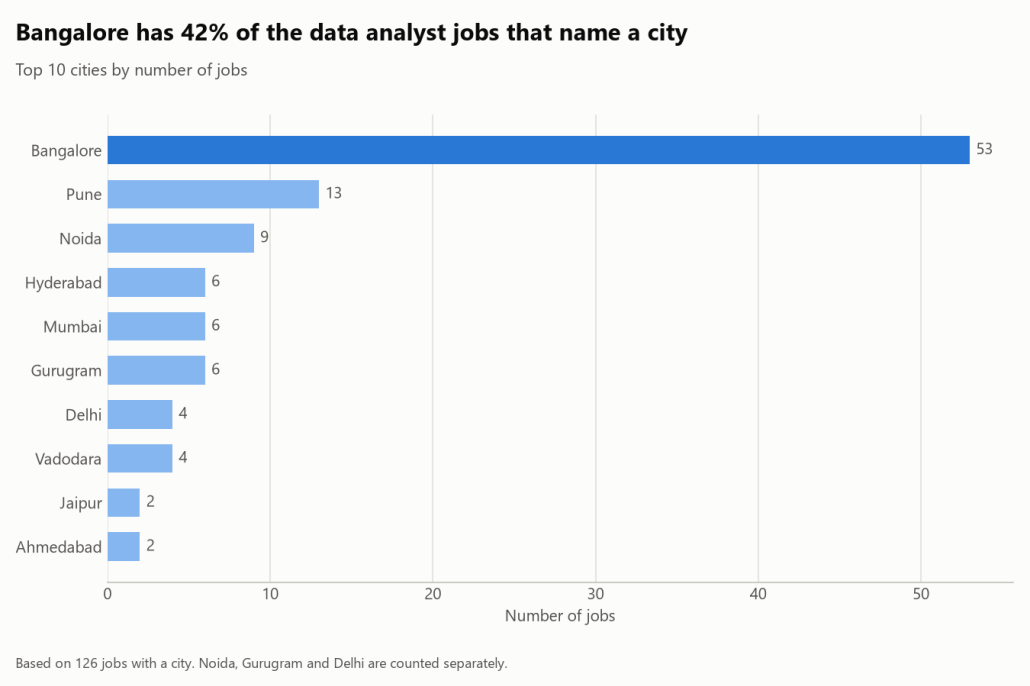


Figure 1. Top 10 cities by number of data analyst jobs (126 jobs with a city).

Bangalore has 53 jobs, about 42% of those that name a city. Pune (13) and Noida (9) follow. Delhi NCR is split into Noida, Gurugram (6) and Delhi (4); added together (19) it is larger than Pune. 35 jobs (22%) only say "India", so their city is unknown.

5.2 Which companies are hiring?

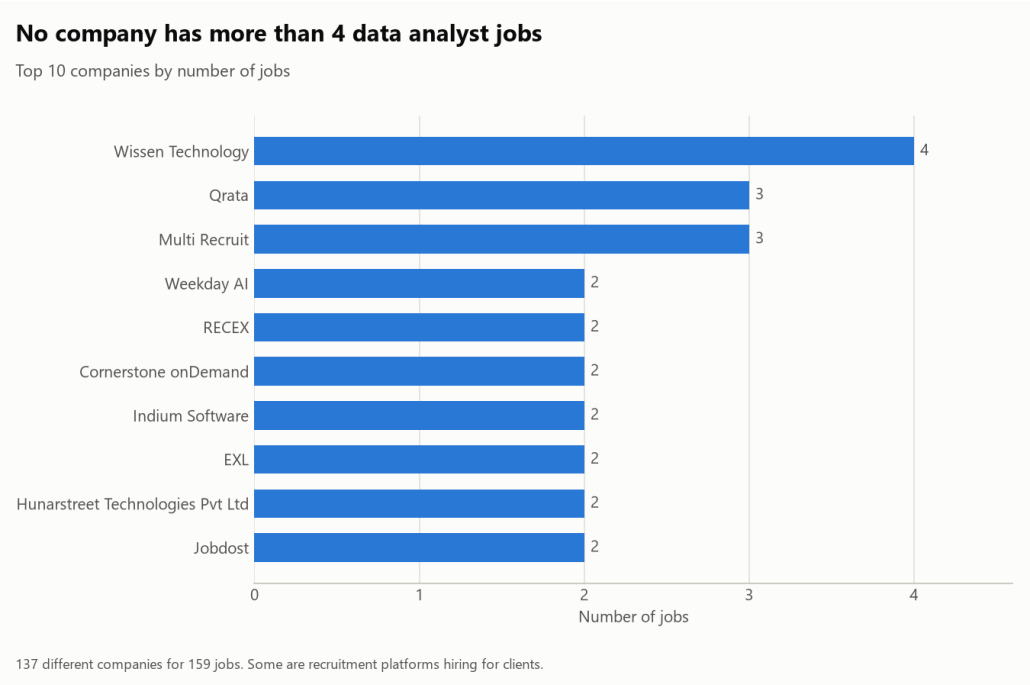


Figure 2. Top 10 companies by number of jobs.

Hiring is very spread out: 137 companies for 159 jobs with a company name. Wissen Technology leads with only 4 jobs, followed by Qrata and Multi Recruit with 3 each. Several names are recruitment platforms (for example Weekday AI posts roles for its clients), so the company shown is not always the final employer.

5.3 Contract and seniority

119 jobs (74%) are full time and the rest do not say; none is part time. Contract type is missing for 150 jobs (93%); of the 11 that say, 7 are permanent and 4 are contract. Only 5 titles show a level (4 senior, 1 junior), so seniority is better studied through years of experience.

5.4 How much experience is asked?

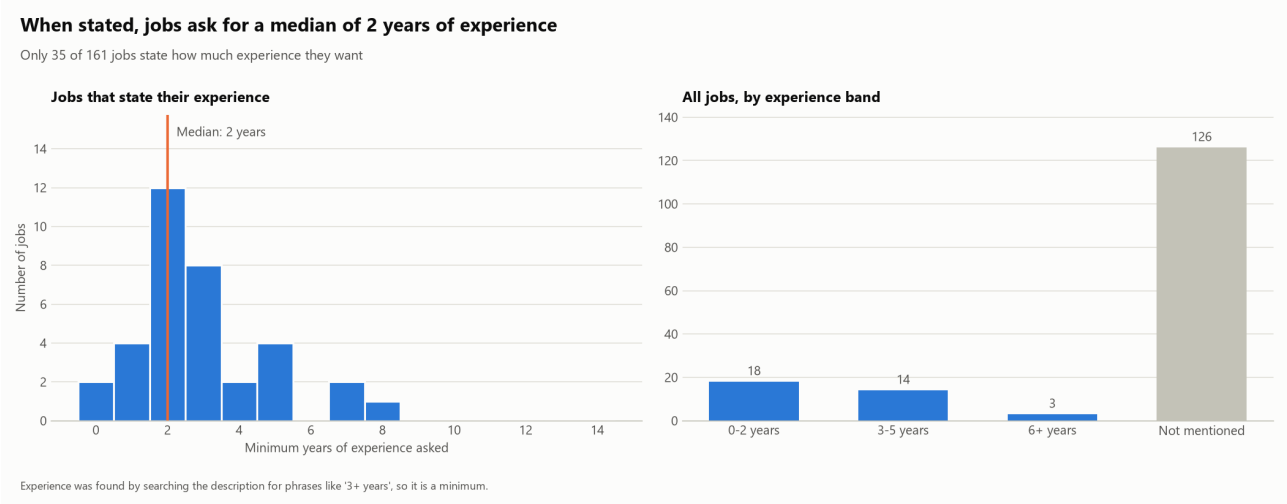


Figure 3. Minimum years of experience asked, and all jobs by experience band.

Experience could be found for only 35 of 161 jobs (22%). Among those, the median minimum is **2 years**: 18 jobs ask for 0-2 years, 14 for 3-5 years and 3 for 6+ (the highest is 8). When stated, the role leans towards entry and junior level, but jobs that state experience may differ from those that do not.

5.5 Freshness and remote work

95 jobs (59%) were posted in 2026 and 53 (33%) in the last 30 days. 35 jobs (22%) come from 2019 to 2022 and are probably old listings that were never removed. Only 7 jobs (about 4%) mention remote, work from home or hybrid; the true share may be slightly higher because descriptions are cut at 500 characters.

5.6 Salary

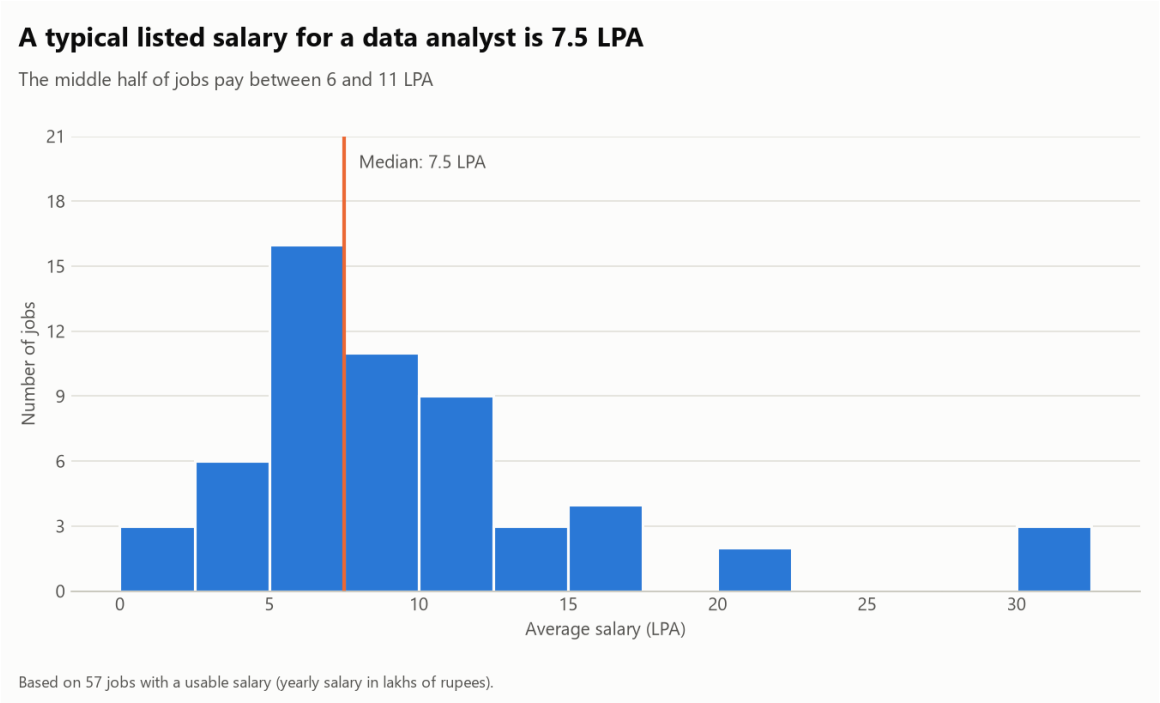


Figure 4. Distribution of average salary (LPA) for 57 jobs with a usable salary.

Only 57 of 161 jobs (35%) have a usable salary. The **median is 7.5 LPA** and the middle half pay between 6 and 11 LPA. The mean (9.7 LPA) is pulled up by a few high salaries (maximum 32.4 LPA), so the median describes a typical job better.

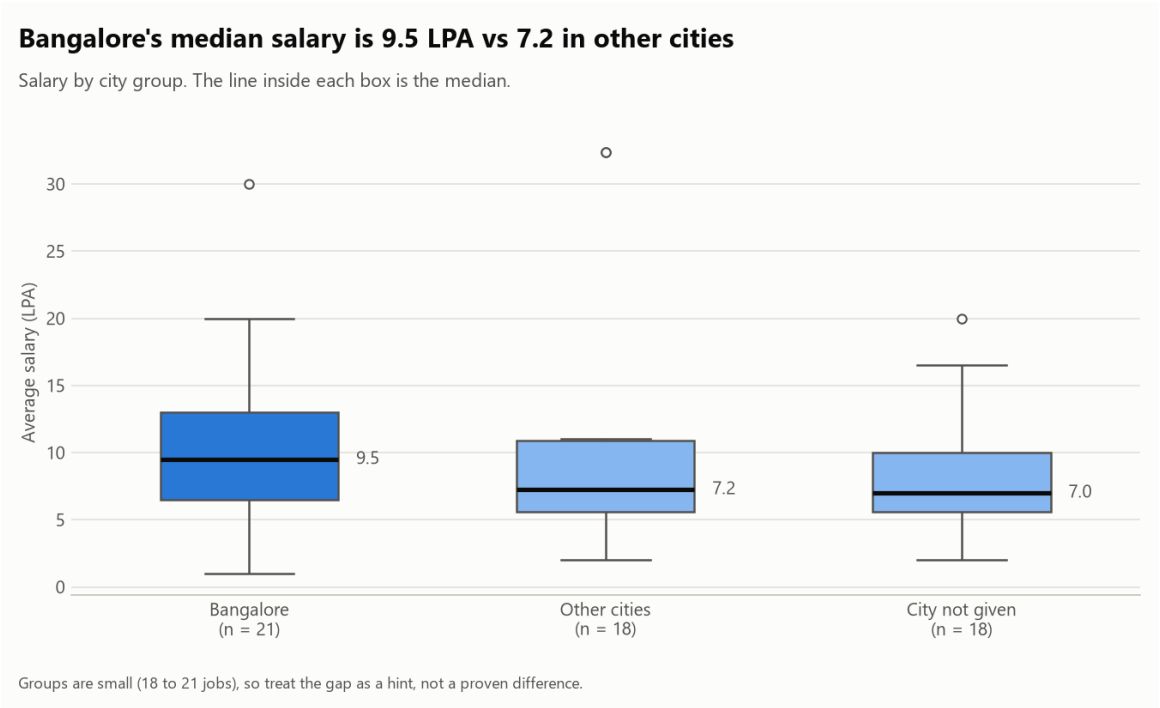


Figure 5. Salary by city group. The line inside each box is the median.

Only Bangalore has enough salary values (21) to stand alone; other cities have 3 or fewer each, so they are grouped. Bangalore's median is **9.5 LPA**, against 7.25 LPA for other cities (18 jobs) and 7.0 LPA where the city is not given (18 jobs). The boxes overlap a lot and the groups are small, so treat the gap as a hint, not a proven difference. Salary and experience could not be compared because very few jobs state both.

5.7 Which tools are asked for most?

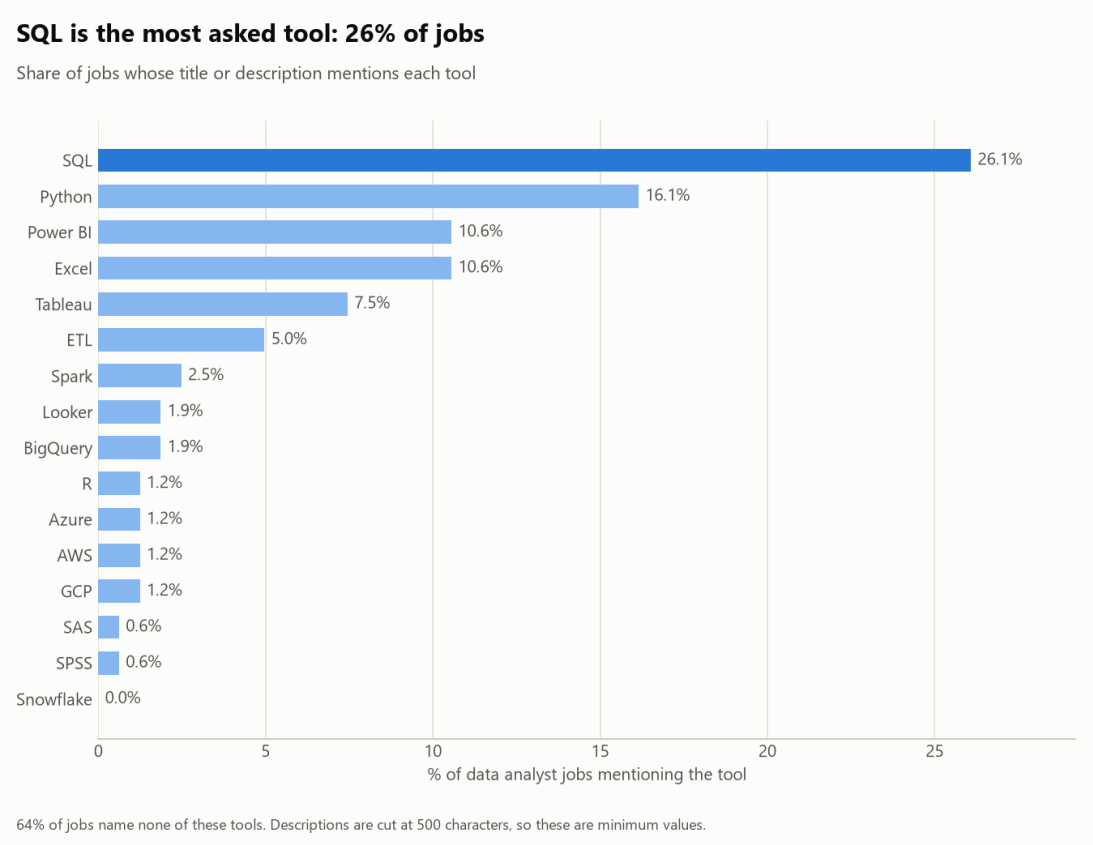


Figure 6. Share of jobs whose title or description mentions each tool.

SQL is the most asked tool (42 jobs, 26%), followed by Python (16%), Power BI and Excel (10.6% each) and Tableau (7.5%). A BI tool (Power BI, Tableau or Looker) appears in 21 jobs (13%). Everything else is at 5% or less. 64% of jobs name none of the 16 tools, mainly because descriptions are cut at 500 characters, so these numbers are **minimums**.

5.8 The nature of the work

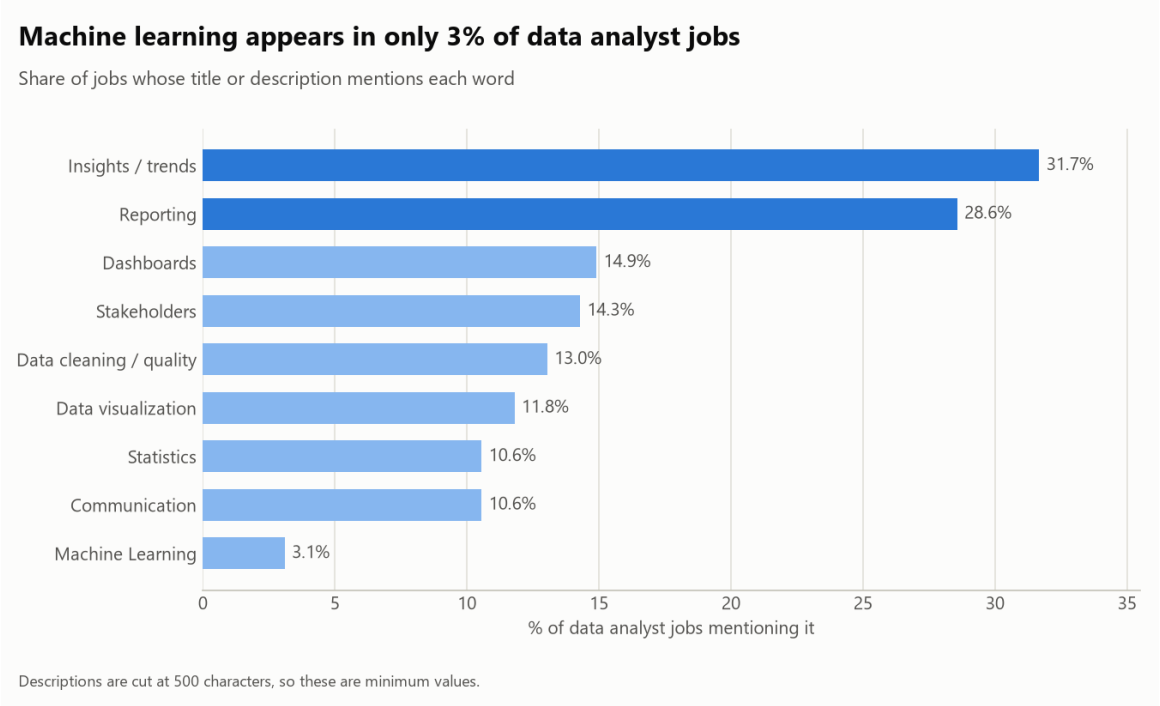


Figure 7. Share of jobs mentioning each concept or soft skill.

Insights/trends (32%) and reporting (29%) lead, then dashboards (15%), stakeholders (14%), data cleaning/quality (13%), visualization (12%), statistics and communication (11% each). Machine learning appears in only 3% of jobs. In this data, the job is mostly about cleaning data, building reports and dashboards, and explaining insights to the business.

5.9 Which tools come together?

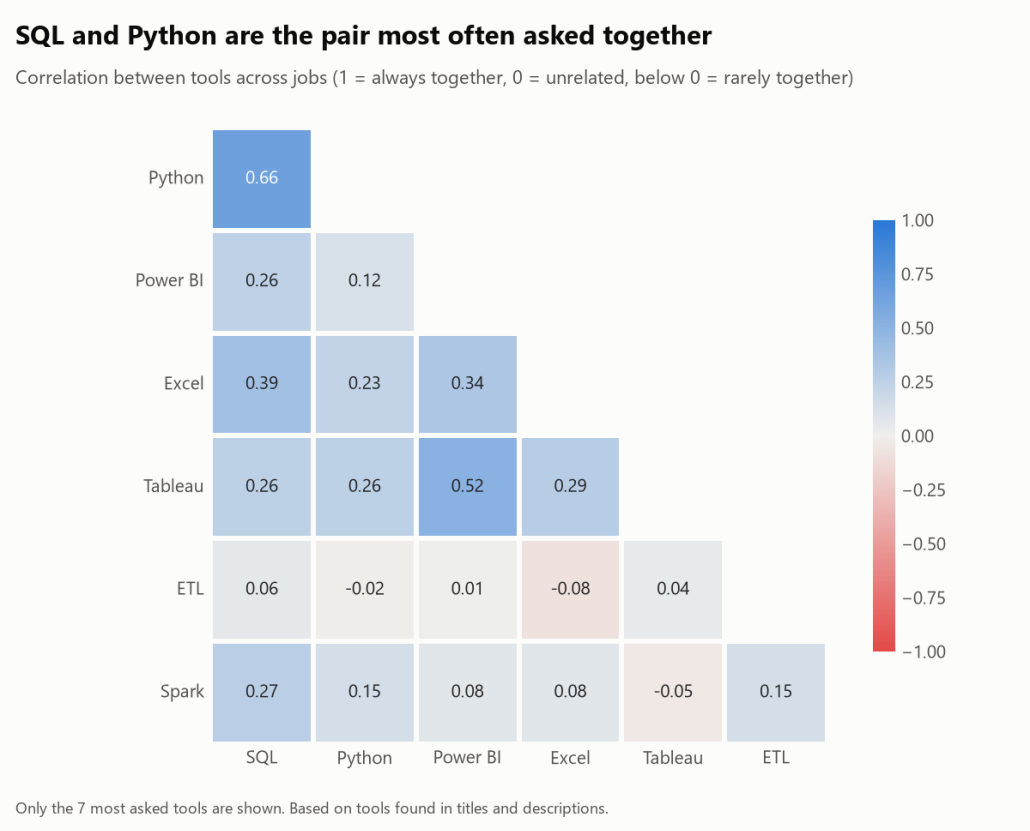


Figure 8. Correlation between the most asked tools across jobs.

- **SQL and Python** are the strongest pair (correlation 0.66): 24 jobs ask for both, 57% of all SQL jobs.
- **Power BI and Tableau** also go together (0.52); 8 jobs ask for both.
- SQL with Excel: 0.39 (13 jobs). SQL with a BI tool appears in 14 jobs.
- The classic SQL + Excel + BI tool combination appears in only 6 jobs.
- ETL and Spark are only weakly linked with the others and point to roles closer to data engineering.

In this sample, SQL is the core and employers add either Python or a BI tool to it.

5.10 Do tools raise salary?

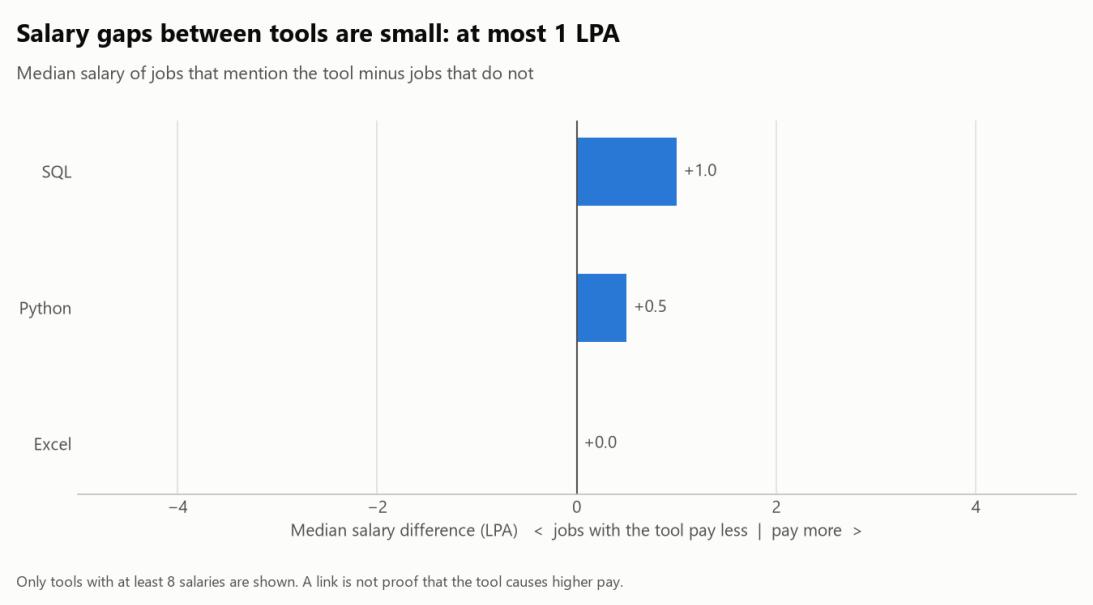


Figure 9. Median salary of jobs mentioning a tool minus jobs that do not (tools with at least 8 salaries).

Jobs that name SQL have a median of 8.5 LPA (20 jobs) against 7.5 LPA for the rest (+1.0). Python gives +0.5 (11 jobs) and Excel +0.0 (8 jobs). Power BI and Tableau have fewer than 8 salaries each and are excluded. All gaps are small, so no tool can be said to raise pay in this sample.

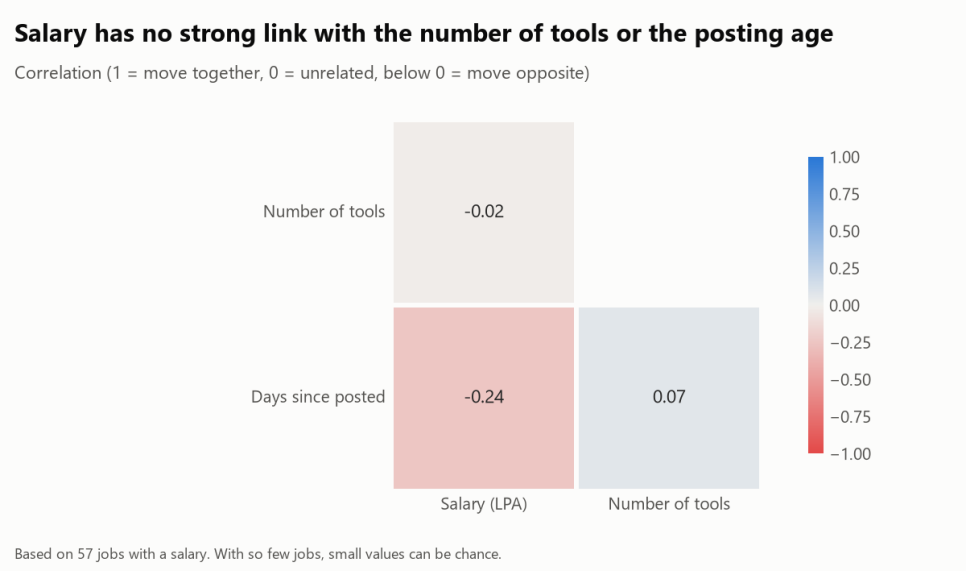


Figure 10. Correlation between salary, number of tools and posting age (57 jobs with a salary).

Salary has no strong link with the number of tools mentioned (-0.02) and only a weak negative link with posting age (-0.24: older postings tend to list slightly lower salaries). With only 57 jobs, small values can be due to chance.

6. Conclusions and recommendations

Question	Answer from this sample
Where / who?	Bangalore leads (42%); Pune and Delhi NCR follow. No company dominates.
Experience?	Median 2 years when stated; 0-5 years covers most jobs.
Pay?	Median 7.5 LPA, middle half 6-11 LPA; Bangalore looks about 2 LPA higher (not proven).
Tools?	SQL first, then Python, Power BI, Excel, Tableau. SQL + Python is the top pair.
Work content?	Insights, reporting and dashboards far above machine learning (3%).
Remote / fresh?	About 4% mention remote; 59% of postings are from 2026.

What this means for a beginner: learn SQL first, then Python (pandas) and one BI tool (Power BI or Tableau). Practise building dashboards and presenting insights, because those words appear in many job posts. Entry-level candidates are a good fit: stated experience is typically 0-3 years.

7. Limitations

- **Small dataset:** 161 jobs and only 57 with a usable salary. Many comparisons use groups of 3 to 20 jobs, so differences can be due to chance.
- **Not the whole market:** the data analyst search stopped at 150 jobs (3 pages of 50).
- **Truncated descriptions:** the API cuts descriptions at 500 characters, so tools, experience and remote work are under-counted. Experience was found for only 22% of jobs.
- **Related roles excluded:** 65 Business Intelligence / BI analyst jobs and 9 Business Analyst jobs were left out on purpose.
- **Old listings:** 22% of postings are from 2019-2022 and may be stale.
- **Keyword matching:** tools and skills were found by searching words, which is simple and can make mistakes.
- **Selection effects:** jobs that state salary or experience may differ from those that do not.

8. Next steps

- Collect more data analyst jobs (more pages and search terms) and re-run the notebook; it reads the new file automatically.
- Change `ROLE_WORD` to `"data analyst|business intelligence|business analyst"` to include related roles and compare them.
- With more data, model salary against experience, city and tools with a simple linear regression.

Project files

File	Purpose
fetch_jobs.py	Collects IT jobs from the Adzuna API and saves the CSV
IT_Jobs_india.ipynb	Selection, cleaning, feature engineering, analysis and charts
requirements.txt	requests, pandas, python-dotenv
.env / .env.example / .gitignore	API credentials template and files kept out of version control
it_jobs_india.csv, data_analyst_jobs.csv, data_analyst_jobs_clean.csv	Raw and cleaned datasets

Note: the notebook also produces charts 03 (contract and level), 05 (year and remote) and 07 (salary vs experience); these were not part of the files supplied for this report, so their results are given in text only.